

Processing Speed -- Guided Notes

Unit tp-1.7 | CompTIA Network+ N10-009 Obj. FC0-U71 1.3 | N10-008 FC0-U71 1.3

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Hz stands for _____ and represents one cycle per second. GHz stands for _____ and equals _____ cycles per second. A CPU running at 3.6 GHz completes approximately _____ billion instruction cycles per second.

Question 2

Clock speed _____ (always/does not always) determine overall CPU performance. A CPU running single-threaded workloads benefits from _____ clock speed, while tasks that can be divided into parallel work benefit from more _____ even at lower clock speeds.

Question 3

A multi-core CPU has multiple _____ on a single chip, each able to execute instructions _____ (sequentially/independently). A 4-core CPU can technically work on _____ different tasks at the same time, improving _____ when running multiple programs simultaneously.

Question 4

CPU cache is ultra-fast memory built directly into the processor. _____ cache is the smallest and fastest, closest to the core. _____ cache is shared within a core and slightly larger. _____ cache is the largest and shared among all cores. Cache reduces how often the CPU must wait for data from _____, which is much slower.

Question 5

Thermal throttling occurs when a CPU reaches its maximum safe _____ and automatically reduces its _____ to generate less heat. While this prevents hardware damage, it results in _____ performance. The long-term fix is to improve _____ or reapply thermal paste.

Question 6

A _____ is a standardized test used to measure and compare CPU, GPU, or system performance. The score allows comparisons between different hardware. However, benchmark results _____ (always/do not always) reflect real-world performance for every type of workload.

Question 7

Hyper-Threading (Intel) and SMT (AMD) allow each physical CPU core to handle _____ threads simultaneously. A 4-core CPU with hyper-threading appears to the OS as _____ logical processors. This improves performance when the CPU has _____ work waiting than physical cores to run it.

Question 8 -- Real World Application

REAL WORLD: A student is choosing between two laptops. Laptop A has a 4-core CPU at 4.0 GHz. Laptop B has an 8-core CPU at 2.8 GHz. The student assumes Laptop A is faster because it has a higher clock speed. For which tasks would Laptop A actually perform better, and for which would Laptop B perform better? Explain your reasoning.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-1-7-processing-speed/tp-1-7-processing-speed-answer-key.pdf
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